

## 50 Java Stream API Practice Problems

### Beginner Level (1-15)

1. Print all even numbers from a list.
2. Print all odd numbers from a list.
3. Convert all strings to uppercase.
4. Convert all strings to lowercase.
5. Find the first element that starts with "A".
6. Find any element that ends with "z".
7. Count elements greater than a given number.
8. Check if any number is divisible by 5.
9. Check if all elements are positive.
10. Remove null values from a list.
11. Remove empty strings from a list.
12. Sort the list in ascending order.
13. Sort the list in descending order.
14. Find the maximum value from a list.
15. Find the minimum value from a list.

### Intermediate Level (16-35)

16. Find sum of all numbers in a list.
17. Find average of numbers in a list.
18. Find product of all numbers.
19. Get a list of square of numbers.
20. Get a list of cube of numbers.
21. Filter names with more than 4 characters.
22. Find second highest number in a list.
23. Find duplicate elements in a list.
24. Remove duplicates from a list.
25. Group strings by their length.
26. Count frequency of each element.
27. Join all strings with comma separator.

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28. Skip first 3 elements and print rest.
29. Limit to first 5 elements and print.
30. Partition list into even and odd numbers.
31. Find the longest string in a list.
32. Find the shortest string in a list.
33. Find the number of characters in each string.
34. Collect result into a Set instead of List.
35. Flatten a list of lists using flatMap.

### Advanced Level (36-50)

36. Group employees by department.
37. Count employees in each department.
38. Find average salary by department.
39. Sort employees by salary.
40. Find highest paid employee.
41. Find youngest and oldest employee.
42. Filter employees whose name starts with "S".
43. Convert a list of objects to a Map.
44. Find duplicate words in a string.
45. Count word frequency in a sentence.
46. Reverse each word in a string using streams.
47. Convert list of strings to a single string with Collectors.joining().
48. Implement pagination using skip and limit.
49. Partition numbers into prime and non-prime.
50. Find common elements between two lists.